CompSci 234/NetSys 210
Advanced Topics in Networking

Winter 2019

Unit 13: Cellular Networks

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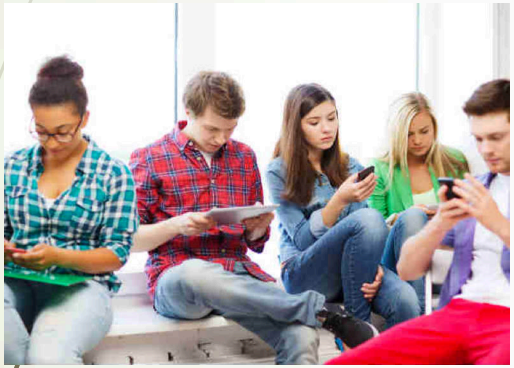
Some Slide adopted from Prof. Chunyi Peng's and Qualcomm's public materials

Agenda

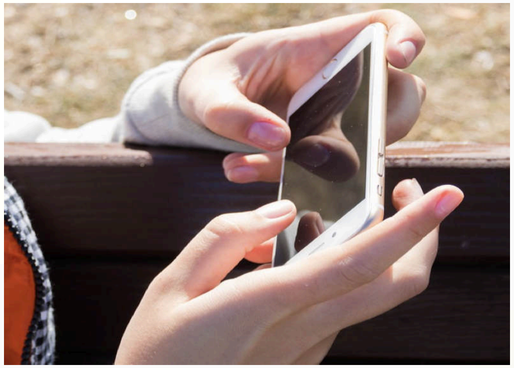
- 5G Cellular Networks
- Evolution of Cellular Networks
- Network Architecture
- Network Operations and Protocol Stacks

Mobile Internet Anywhere Anytime

- Smartphone is the most disruptive technology in the past decade!

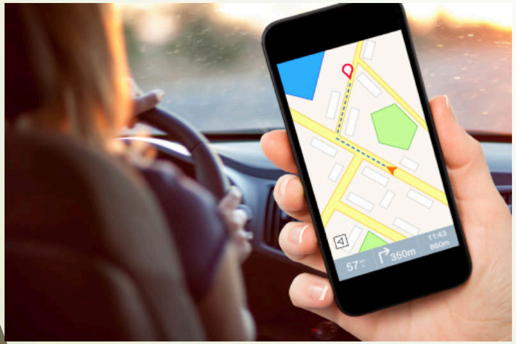


Indoor

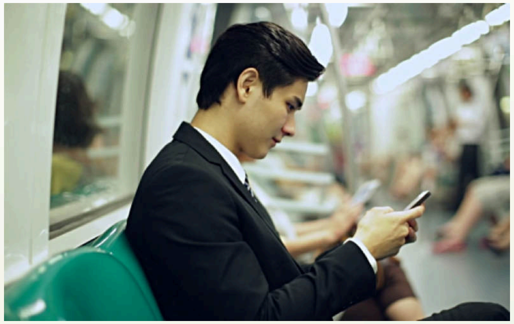


Outdoor

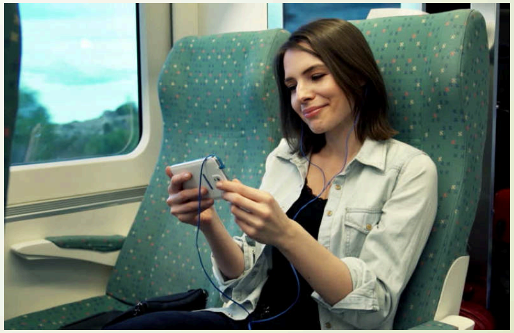
Walking



Driving

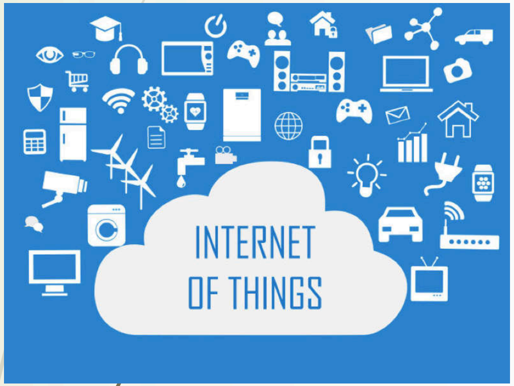


Subway



High-speed train

Future 5G Applications



Internet of Things

Tactile Internet

Autonomous Driving

AR/VR

Remote Healthcare

Drone-based Delivery

Wishlist of 5G Cellular Networks

- Critical requirements
  - High bandwidth
  - Low latency
  - High efficiency
  - High reliability
  - Good security

4G
2010s

5G
2020s

Faster and Lower Latency
10Gps peak rate
< 1ms latency

Better-connected
10000x traffic
1000x bandwidth
10-100x devices

Higher mobility
300+ Kmh

Ultra-reliable
99.999%

Energy-efficient
...

Agenda

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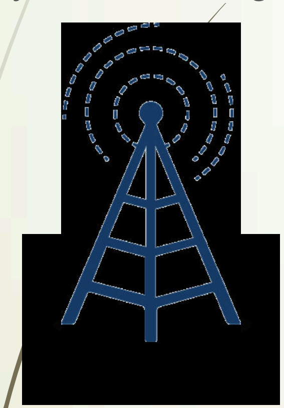
1G Cellular Networks: the Foundation

1

Licensed Spectrum

Cleared spectrum for exclusive use by mobile technologies

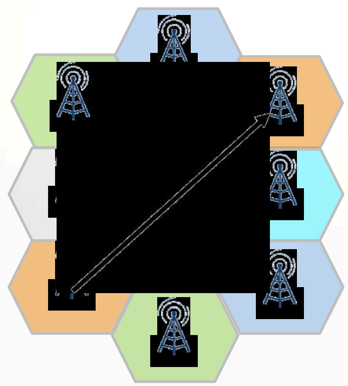


Operator-deployed **base stations** provide access for subscribers

2

Frequency Reuse

Reusing frequencies without interference through geographical separation

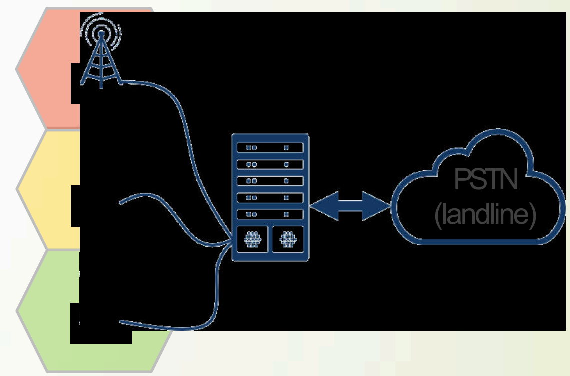


Neighboring **cells** operate on different frequencies to avoid interference

3

Mobile Network

Coordinated network for seamless access and seamless mobility



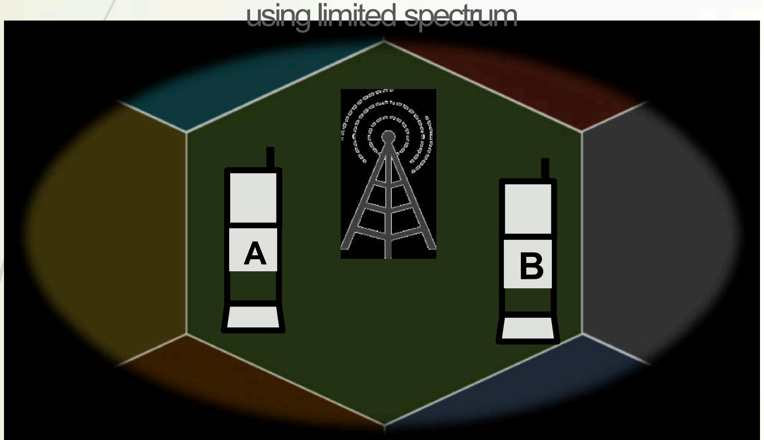
Integrated, transparent **backhaul network** provides seamless access

Q: What's the single key concept in cellular network?

1G Offers Analog Voice Service with Some Limitations

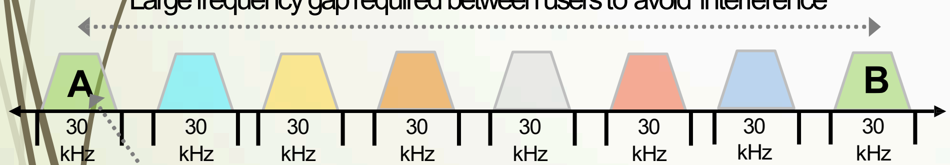
Limited Capacity

Analog transmissions are inefficient at using limited spectrum



Frequency Division Multiple Access (FDMA)*

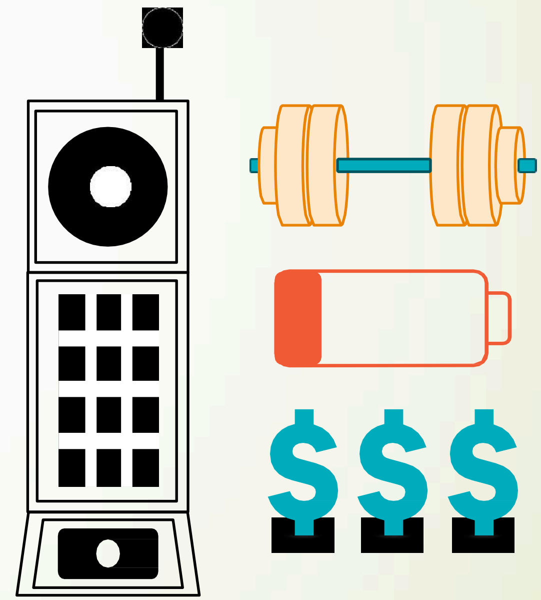
Large frequency gap required between users to avoid interference



Support for only 1 user (analog phone call) per channel

Limited Scalability

Analog devices are large/heavy, power inefficient, and high cost



* Example shown based on AMPS 1G technology